PROJECT REPORT

AI Study Companion for Engineering Exams

An AI-powered learning assistant to help engineering students prepare smarter for examinations.

Project Title	AI Study Companion for Engineering Exams
Project Type	Artificial Intelligence / Educational Technology
Prepared By	Student Project
Institution	College / University
Academic Year	2026–2027

Submitted in partial fulfilment of academic project requirements.

Abstract

Engineering students often need to study large syllabi, understand difficult concepts, revise quickly, and practise exam-style questions. This project proposes an AI Study Companion for Engineering Exams: a digital learning assistant that supports personalised preparation through topic explanations, summaries, flashcards, practice questions, and revision planning. Students can enter a subject or topic, choose their preferred level of detail, and receive structured study material. The system can also help organise revision based on exam dates and topics that need more attention. The companion is intended to support learning rather than replace textbooks, lecturers, or official course materials. Its usefulness depends on the quality of the supplied content and the accuracy of the AI model; therefore, students should verify important answers against trusted academic sources.

Keywords: Artificial Intelligence, Engineering Education, Personalised Learning, Exam Preparation, Study Assistant.

1. Introduction

Engineering courses include theoretical concepts, mathematical problem-solving, programming, laboratory work, and application-based questions. Students may struggle to identify important topics, revise consistently, or find explanations suited to their learning level. A study companion powered by artificial intelligence can make preparation more organised by turning topics into digestible explanations and practice activities. This project focuses on a student-friendly interface that brings common study tasks into one place.

2. Problem Statement

Students commonly rely on scattered notes, textbooks, websites, and separate quiz tools. This can make revision time-consuming and difficult to personalise. The proposed system aims to provide a single assistant for explaining topics, generating revision resources, practising questions, and planning study sessions while reminding users to verify AI-generated content.

3. Objectives

- Provide clear, topic-focused explanations for engineering subjects.
- Generate concise summaries, revision notes, and flashcards.
- Create practice questions and quizzes with answer explanations.
- Help students build a study plan based on subjects, available time, and exam dates.
- Track completed topics and quiz performance to guide revision.
- Encourage responsible AI use by showing limitations and recommending source verification.

4. Proposed System

The system accepts a subject, topic, question, or uploaded/entered study text. It processes the request using an AI language model and presents a structured response. For learning activities, it can generate a summary, key definitions, worked-example outlines, flashcards, or a quiz. A planner stores the student’s selected subjects and target dates, while a progress module records completed activities and quiz scores.

Typical workflow

- Student selects a subject and enters a topic or question.
- The system identifies the requested task, such as explanation, summary, or quiz.
- The AI service generates a response in a selected format and difficulty level.
- The student reviews the response, practises questions, and checks answers.

- Progress is saved and used to suggest future revision topics.

5. Main Modules

5.1 AI Topic Explainer

Explains technical concepts in simple language, with definitions, step-by-step reasoning, and examples where appropriate.

5.2 Notes and Summary Generator

Converts supplied notes or a topic outline into concise revision notes, key points, and formula lists. It should preserve important technical details.

5.3 Quiz and Flashcard Generator

Creates multiple-choice, short-answer, and flashcard-style practice activities. Answer explanations help students understand mistakes.

5.4 Study Planner

Lets students enter subjects, exam dates, and available study hours. It produces a practical schedule that can be adjusted by the student.

5.5 Progress Tracker

Records completed topics, quiz results, and revision status, then highlights areas that may need more practice.

6. Suggested Technology Stack

Layer	Suggested technology	Purpose
Frontend	HTML, CSS, JavaScript or React	Student interface and study dashboard
Backend	Python with Flask or FastAPI	Request handling and application logic
AI service	An approved LLM API or locally hosted model	Explanations, summaries, and question generation
Database	SQLite for prototype; PostgreSQL for deployment	Users, plans, quiz results, and progress
Deployment	College server or a trusted cloud platform	Hosting and access

7. Methodology

- Requirement analysis: identify student needs, target subjects, and core features.
- System design: define screens, data flow, AI prompts, and storage requirements.
- Prototype development: build the interface, backend endpoints, and AI integration.
- Testing: check usability, response relevance, quiz correctness, and error handling.
- Evaluation: gather student feedback and measure task completion, usefulness, and response quality.
- Improvement: refine prompts, add source references where available, and correct recurring issues.

8. Functional and Non-Functional Requirements

Functional requirements: topic input, explanation generation, notes creation, quiz/flashcard generation, study-plan creation, and progress tracking.

Non-functional requirements: usable interface, reasonable response time, secure handling of student data, reliability, accessibility, and maintainability.

9. Benefits

- Saves time when organising notes and revision activities.
- Offers explanations in different levels of detail.
- Encourages active recall through quizzes and flashcards.
- Helps students organise preparation across multiple subjects.
- Makes revision more consistent by tracking progress.

10. Limitations and Responsible Use

AI-generated answers can contain errors, omit syllabus-specific details, or provide incorrect calculations. The prototype should not be treated as an authoritative source. Students should cross-check formulas, derivations, and factual claims with textbooks, lecture notes, and faculty guidance. Uploaded materials and personal data should be handled securely, with clear consent and minimal data collection. The system should not claim to predict exam questions with certainty.

11. Future Enhancements

- Support for syllabus and past-paper uploads with source-linked answers.
- A formula reference library and step-by-step problem-solving mode.
- Speech input and text-to-speech for accessibility.
- A mobile-friendly application and offline access to saved notes.
- More detailed analytics to identify weak topics over time.
- Integration with a college learning management system, if approved.

12. Conclusion

The AI Study Companion for Engineering Exams is a proposed educational tool that combines topic explanations, revision resources, practice questions, planning, and progress tracking. By helping students structure their preparation and practise actively, it can make study sessions more focused and manageable. A successful implementation should prioritise academic accuracy, student privacy, usability, and transparency about AI limitations. The system is best used as a companion to formal teaching and trusted course materials.

References / Further Reading

- Course textbooks, lecture notes, and official syllabus documents for the relevant engineering subjects.
- Documentation for the selected programming framework, database, and AI model/API.
- Institutional guidelines on academic integrity, privacy, and responsible AI use.